

6. (a) Fertilisers usually contain compounds of three essential elements required for healthy and productive plant growth. The table shows the properties of two nitrogenous fertilisers.

Property	Ammonium nitrate	Urea
% nitrogen	34	46
Mass of fertiliser absorbed by plants per 2000 litre spreader (kg)	690	560
Solubility	very soluble	very soluble
Cost (£/tonne)	360	350
Loss of ammonia due to evaporation	low	high
Weather dependency	effective in all conditions	effectiveness dependent on specific temperature and rainfall conditions

- (i) Which fertiliser leads to the absorption of more **nitrogen** from a full 2000 litre spreader? Show your working. [2]

Fertiliser

- (ii) Urea is the most widely used fertiliser in the world. However, many British farmers prefer to use ammonium nitrate.

Put a tick (✓) in the box next to the statement which suggests the **main** reason for this. [1]

ammonium nitrate contains less nitrogen than urea

☐

ammonium nitrate is better suited to British weather conditions than urea

☐

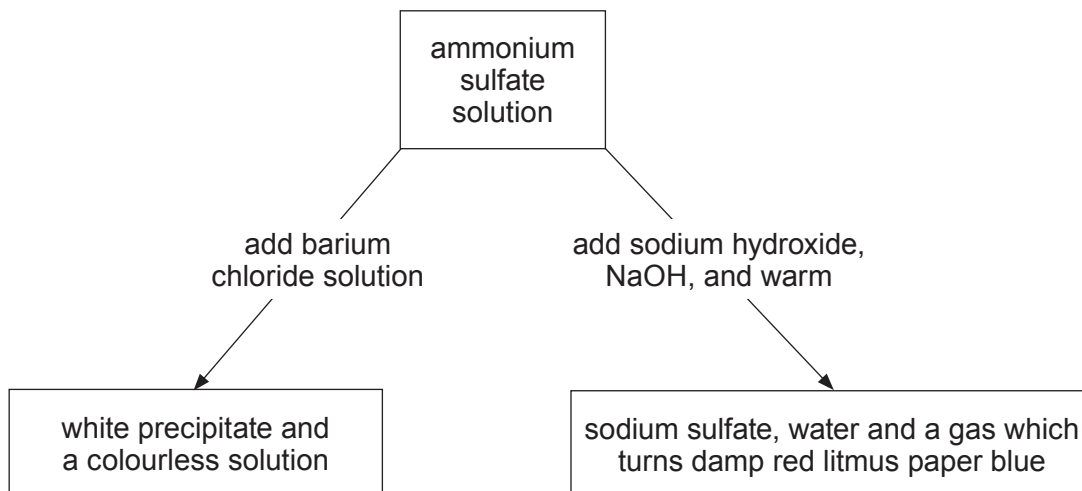
more ammonium nitrate is absorbed by plants than urea

☐

ammonium nitrate is more expensive than urea

☐


- (b) The flow chart shows the chemical tests for the identification of the ions present in ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$.



- (i) Write a balanced **symbol** equation for the reaction between ammonium sulfate and sodium hydroxide. [2]

- (ii) The symbol equation below represents the reaction occurring between ammonium sulfate solution and barium chloride solution.



Write the **ionic** equation for the reaction. Include the state symbols. [2]

..... + $\longrightarrow$

